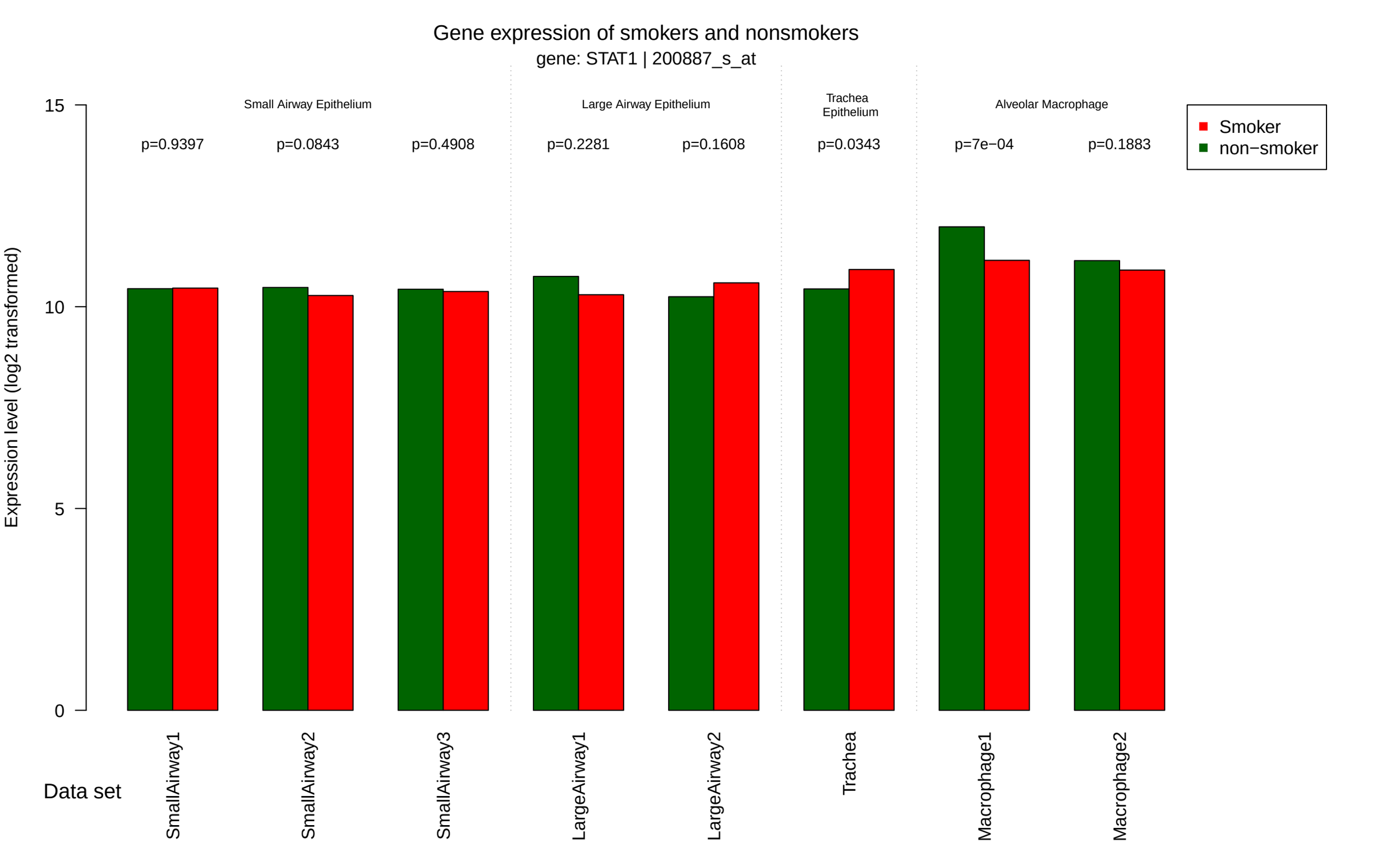
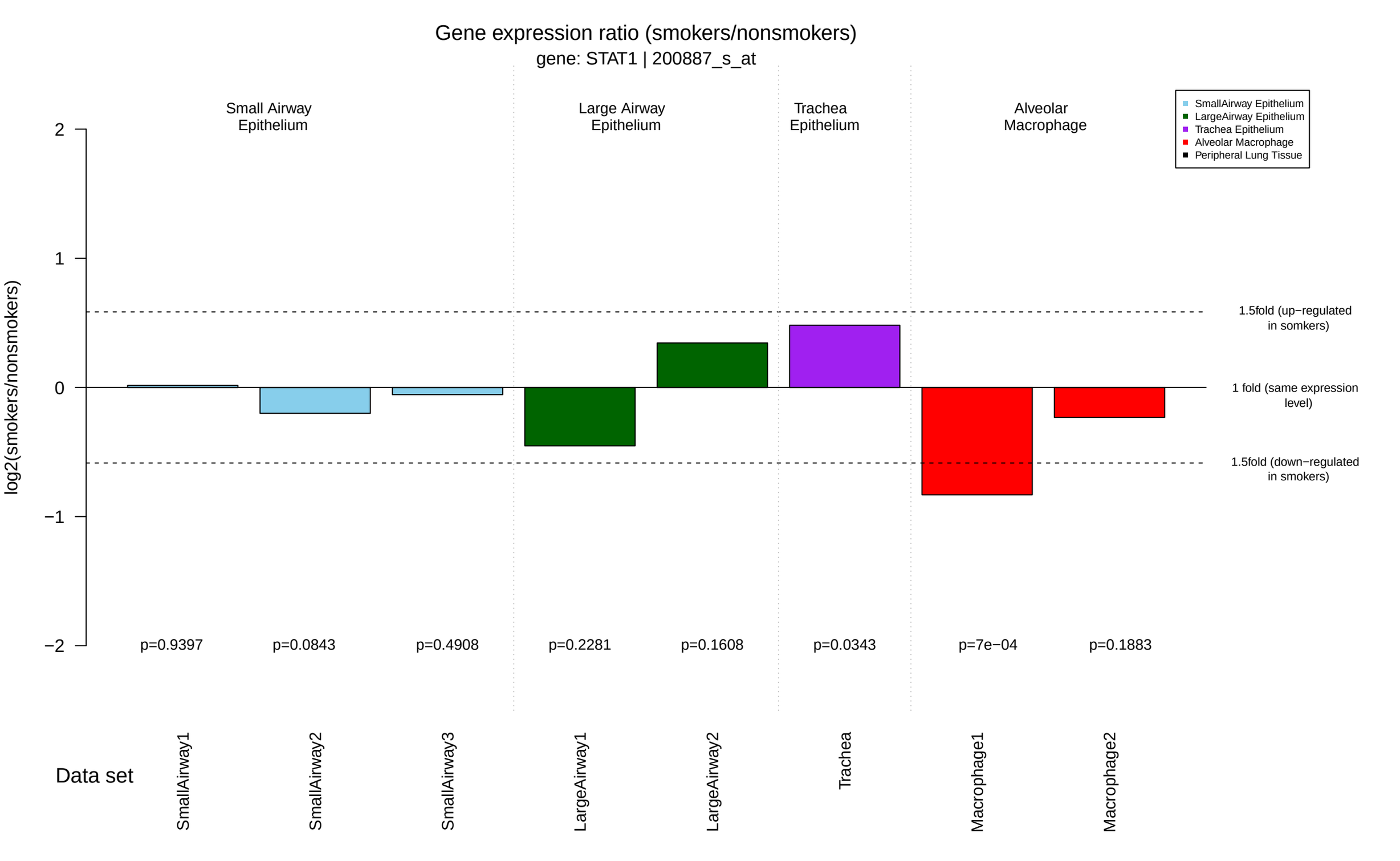
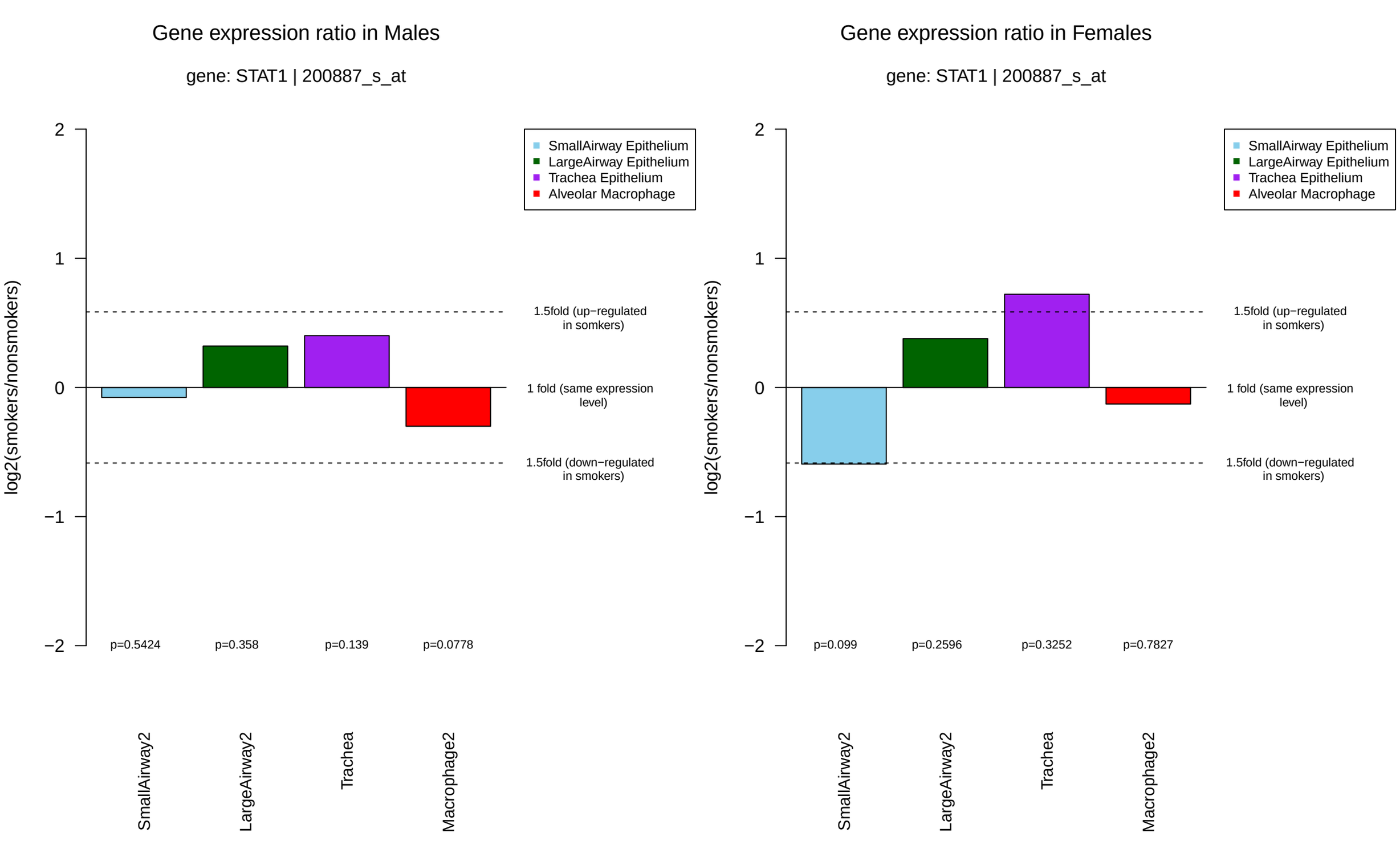
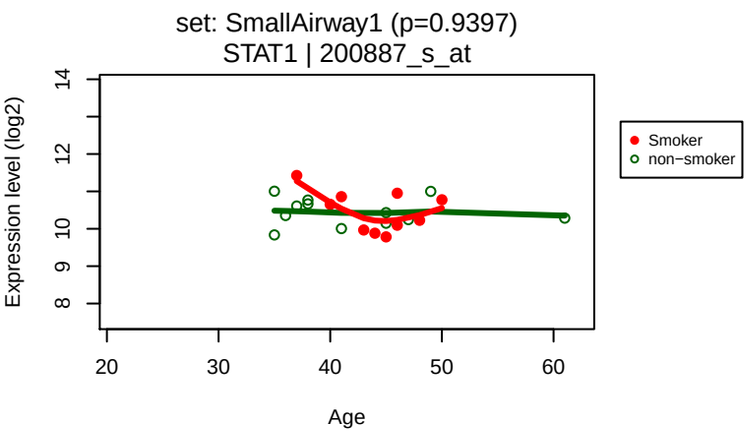
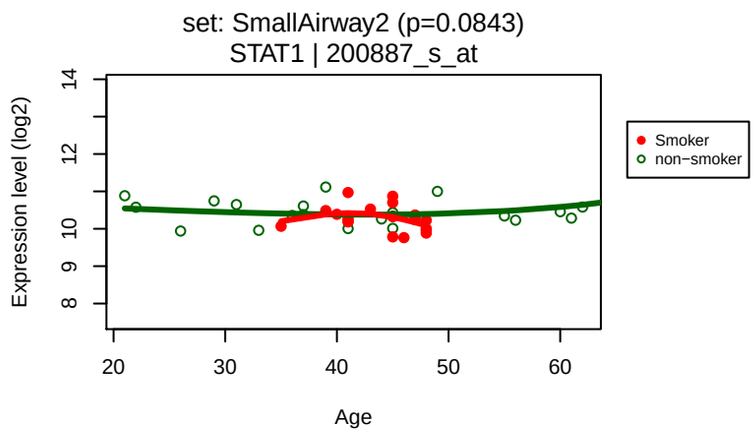
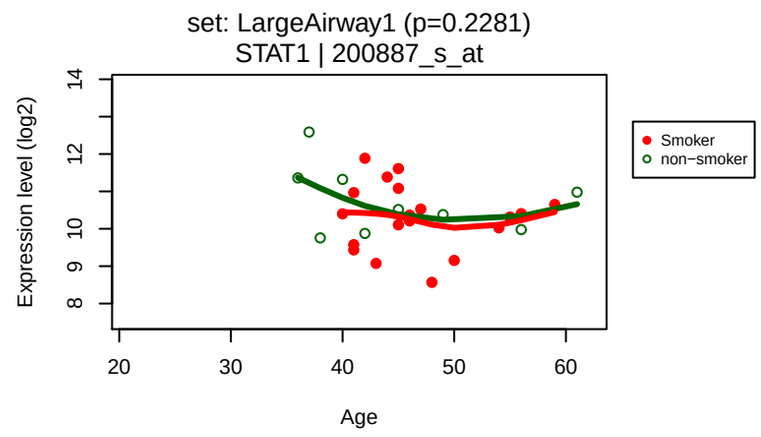

Gene expression of smokers and nonsmokers
gene: STAT1 | 200887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9397
p=0.0843
p=0.4908
p=0.2281
p=0.1608
p=0.0343
p=7e−04
p=0.1883
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

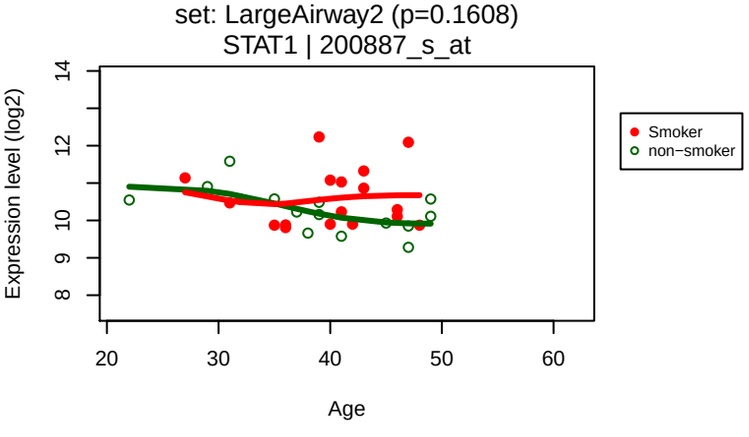

Gene expression ratio (smokers/nonsmokers)
gene: STAT1 | 200887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9397
p=0.0843
p=0.4908
p=0.2281
p=0.1608
p=0.0343
p=7e−04
p=0.1883
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

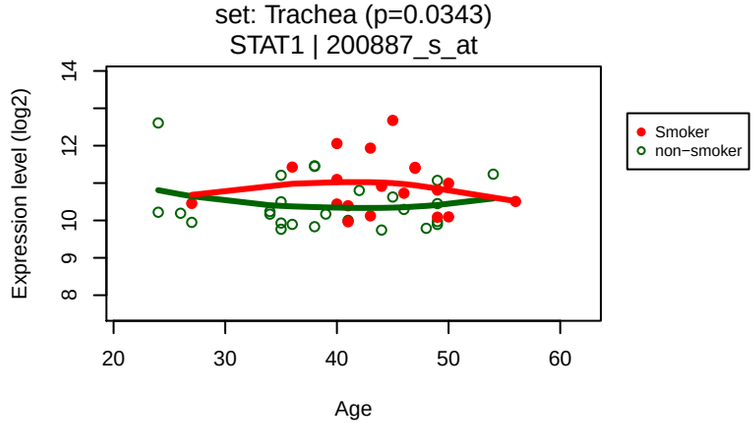

Gene expression ratio in Males
gene: STAT1 | 200887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5424
p=0.358
p=0.139
p=0.0778
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STAT1 | 200887_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.099
p=0.2596
p=0.3252
p=0.7827
SmallAirway2
LargeAirway2
Trachea
Macrophage2

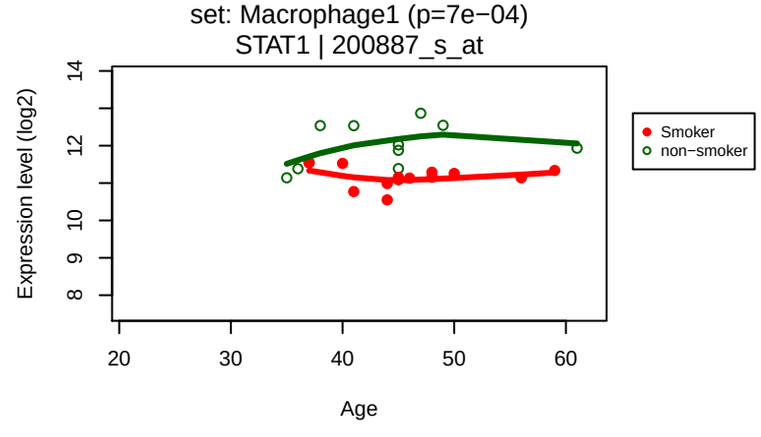

set: SmallAirway1 (p=0.9397)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker

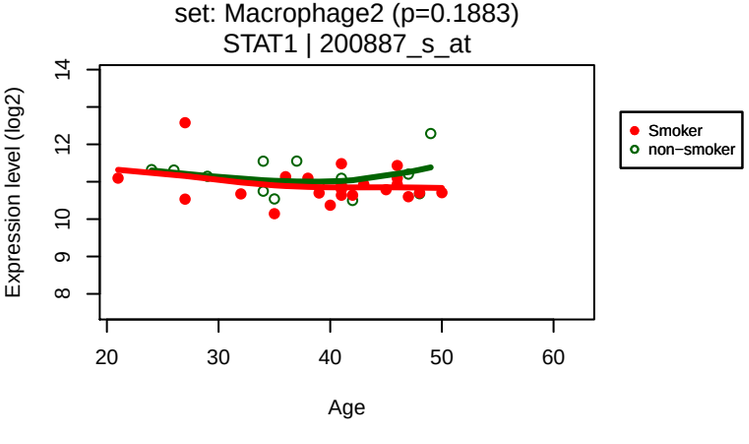

set: SmallAirway2 (p=0.0843)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker

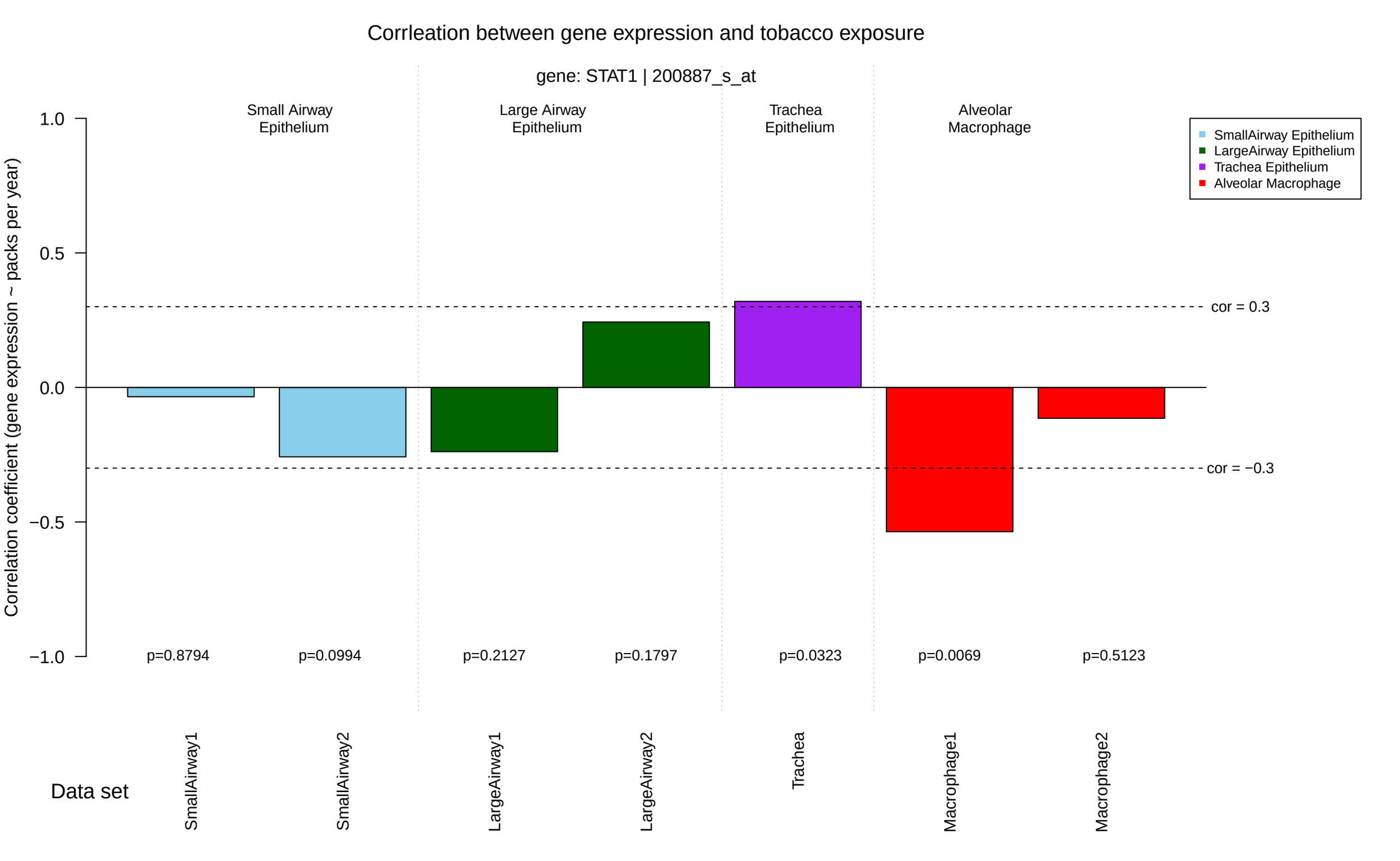

set: LargeAirway1 (p=0.2281)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1608)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0343)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=7e−04)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1883)
STAT1 | 200887_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STAT1 | 200887_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8794
p=0.0994
p=0.2127
p=0.1797
p=0.0323
p=0.0069
p=0.5123
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2